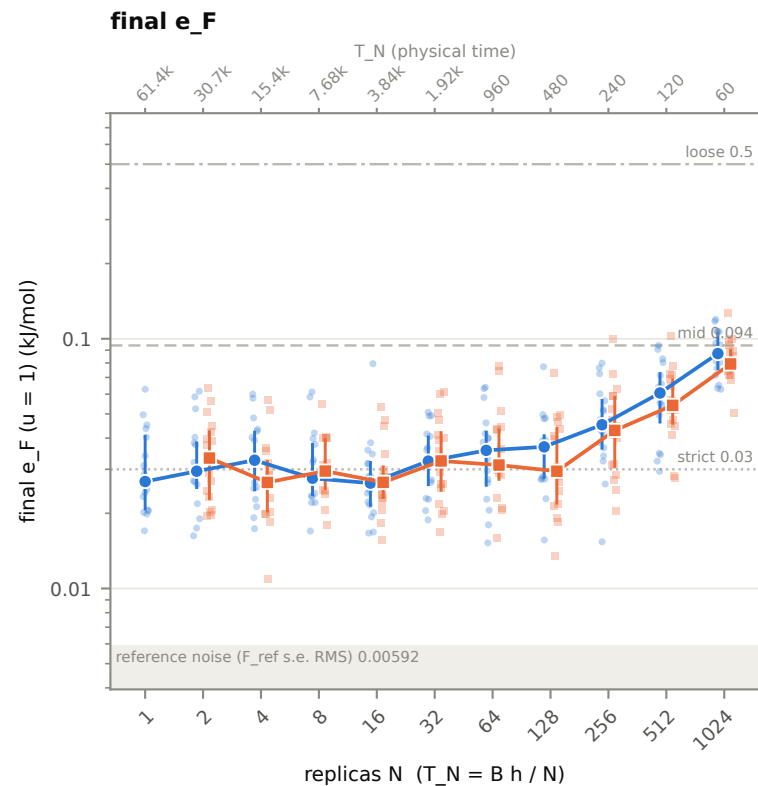
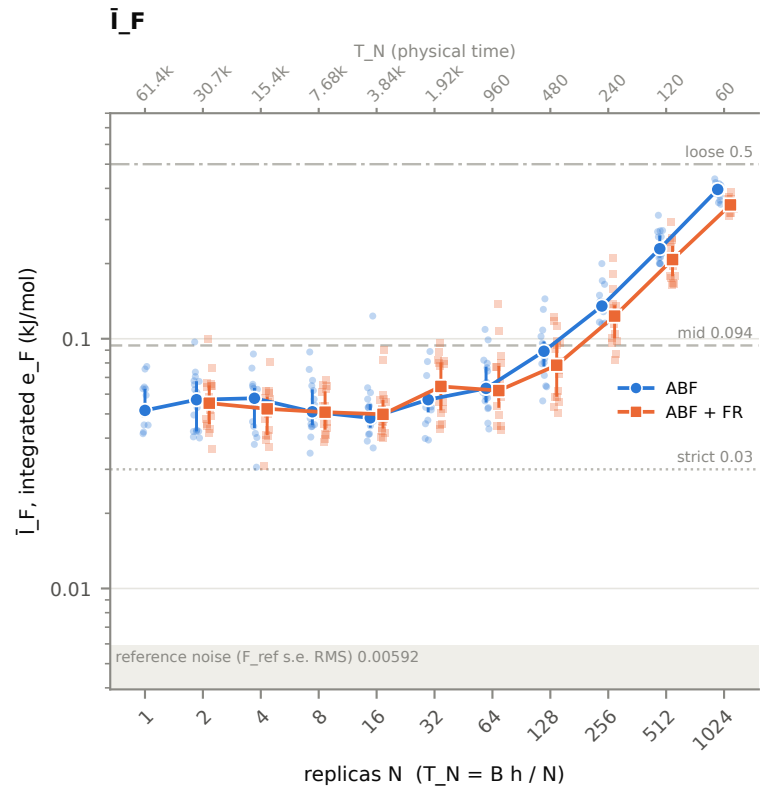


# ethane/LTA 300 K: absolute free-energy error vs N

Equal total budget per arm per seed:  $B = N \times n\_steps = 3.072e+08$  walker-steps at every  $N$  ( $h = 0.0002$ ,  $T\_N = B \cdot h / N$ ); 16 seeds per ( $N$ , method). ABF and ABF + FR: seed median, IQR and per-seed points.



ethane/LTA 300 K · S1\_free\_energy\_abs\_vs\_N · status: all planned (N, seed, method) complete · FR is not defined at  $N = 1$  (ABF only, by design) · LTA  $N < 50$ : FR uses the finite-N cap extension  $\max(1, \text{floor}(0.02 \cdot N))$ .  
Markers: seed median; whiskers: IQR over seeds; faint points: individual seeds. Grey lines: the frozen strict / mid / loose thresholds of  $e(t)$  (shown for scale on the integrated errors as well).  
Source results/equal\_budget\_v2/lta\_T300/analysis/summary.json (generated 2026-10-10T07:06:16Z, eqb\_metrics/2, 10000 bootstrap resamples); plot\_synthesis/2.